MARMARA UNIVERSITY
Faculty of Engineering — Dept. of Electrical and Electronics Engineering

Project Proposal Form

Courses: EE4084 - Kalman and Bayesian Filters
EEE7023 - Advanced Signal Processing

Submission Deadline: 22/04/2026 @ 23:55

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1 Project Identity

Proposed Title: ____
(Select from the curated list in ***Appendix A*** *or propose a custom title.)*

2 Motivation & Problem Statement

Describe the engineering significance of the problem. Why are Bayesian techniques required? Refer to [1, 2] for theoretical grounding.

3 Mathematical Framework

3.1 Process and Measurement Models

Define your state-vector x_k and the observation z_k:

- **Evolution:** $x_k = f(x_{k-1}, u_{k-1}) + w_{k-1}, \quad w_k \sim \mathcal{N}(0, Q)$
- **Observation:** $z_k = h(x_k) + v_k, \quad v_k \sim \mathcal{N}(0, R)$

4 Data Acquisition and Generation Strategy

Explicitly state how you will obtain or generate the data for your filter.

- **Ground Truth Generation:** Explain how you will generate the "true" state $x_{0:K}$ (e.g., using a specific kinematic motion model or a high-fidelity reference signal).
- **Measurement Simulation:** Describe the process of generating $z_{1:K}$. Specify the sampling frequency (f_s), the targeted Signal-to-Noise Ratio (SNR), and how you will model measurement artifacts (e.g., outliers, dropouts, or non-Gaussian multipath).
- **External Datasets (if applicable):** Identify any public datasets to be used (e.g., KITTI for SLAM, PhysioNet for PPG) and describe the pre-processing steps required.

5 Algorithmic Approach & Software Strategy

Specify your choice of filter (EKF, UKF, PF, or Hybrid) and justify it. List the primary tools/libraries (MATLAB, Python/NumPy, etc.) you will employ.

6 Milestones

Week	Primary Objective
W1-2	Mathematical derivation and Jacobian/Sigma Point definitions.
W3-5	Development of the Data Generation script and initial filter implementation.
W6-7	Monte Carlo analysis (RMSE/NEES) and final reporting.

Instructor Approval
Prof. Dr. Engin Masazade

A Detailed Project Selection Guide

A.1 Kinematic Tracking and Navigation

- **Autonomous Vehicle Lane Following (EKF):** Fusion of IMU and Camera/GPS sensors.
- **Quadrotor State Estimation (UKF):** High-fidelity attitude and position estimation using rotational dynamics [2].
- **EKF-SLAM for Robotics:** Recursive estimation of robot pose and map landmarks [3].

A.2 Communication Theory and Signal Processing

- **ISAC for 6G Target Tracking:** Tracking targets using bistatic Delay/Doppler reflections from OFDM waveforms.
- **Carrier Phase and Frequency Offset Tracking:** PLL implementation via EKF for QAM receivers.
- **Dynamic Direction-of-Arrival (DOA) Tracking:** Tracking angular movement via antenna arrays.

A.3 Hybrid and Advanced Methodologies (Recommended for EEE7023)

- **KalmanNet (Deep Learning + KF):** Learning the Kalman Gain from data in partially-known dynamics [4].
- **Particle Filter for Non-Gaussian Tracking:** Handling multi-path/heavy-tailed noise in indoor localization [1].
- **Sparse Signal Tracking:** Applying Iterated EKF to track time-varying sparse signals.

References

[1] M. Arulampalam, S. Maskell, N. Gordon, and T. Clapp, "A tutorial on particle filters for online nonlinear/non-Gaussian Bayesian tracking," *IEEE Transactions on Signal Processing*, vol. 50, no. 2, pp. 174–188, 2002.

[2] S. Julier and J. Uhlmann, "Unscented filtering and nonlinear estimation," *Proceedings of the IEEE*, vol. 92, no. 3, pp. 401–422, 2004.

[3] H. Durrant-Whyte and T. Bailey, "Simultaneous localization and mapping: part i," *IEEE Robotics Automation Magazine*, vol. 13, no. 2, pp. 99–110, 2006.

[4] G. Revach, N. Shlezinger, X. Ni, A. L. Escoriza, R. J. G. van Sloun, and Y. C. Eldar, "Kalmannet: Neural network aided kalman filtering for partially known dynamics," *IEEE Transactions on Signal Processing*, vol. 70, pp. 1532–1547, 2022.